A Unified Deep Learning-Based Framework for Anticipatory Driver Drowsiness Detection Leveraging the Composite Drowsiness Score

Vikul J. Pawar[1][a], Kailash D. Kharat[2][b], Suraj R. Pardeshi[3][c], Vaibhav M. Mokale[1][d] and Vijayshri A. Injamuri[1][e]

[1]*Department of Computer Science and Engineering, Government College of Engineering, Aurangabad, India*
[2]*CSMSS Chatrapati Shahu College of Engineering, Chatrapati Sambhajinagar, Maharashtra, India*
[3]*Department of MCA, Government College of Engineering, Aurangabad, India*
vikul.pawar@geca.ac.in, kailashdkharat@gmail.com, surajrp@geca.ac.in, vaibhavmokale21@gmail.com, vijayshri.injamuri@geca.ac.in

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Abstract: One of the leading causes of highway accidents all over the world is driver drowsiness. To resolve this problem, this paper will present a framework of proactive drowsiness detection and avoidance of accidents using deep learning technology. In the model, the spatial, temporal, and contextual features are integrated by a hybrid architecture that is built on Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Vision Transformers (ViTs), as learning frameworks. CNNs identify important spatial features of each frame and LSTMs learn to represent fatigue changing with time. ViTs improve performance through attention mechanisms used to highlight important behavioral indicators such as long duration of eye closure, frequency of yawning, and movement of the head. These complementary features are combined to produce a dynamic drowsiness measurement of the level of alertness of the driver. The framework is tested on the NTHU Drowsy Driver Detection Dataset and real-life driving videos in different light and environmental conditions. The proposed CNN-LSTM-ViT model outperforms standalone CNN and LSTM models with precision of 98.7 and recall of 98.9. The framework has low latency rate which makes it appealing in real-time Advanced Driver Assistance Systems.

1 INTRODUCTION

Your Driver fatigue and drowsiness remain one of the most dangerous risk factors that contribute to road accidents and deaths in the world. The World Health Organization (WHO) and the National Highway Traffic Safety Administration (NHTSA) (Essahraui et al., 2025, Saleem 2022) estimate that almost 20-25 percent of all road accidents in the world can be traced to driver fatigue or micro-sleep episodes, which is especially high in cases of long or boring time on the road. When a driver fails to maintain adequate levels of awareness, their reaction time, cognitive function, and motor control all deteriorate, which result in a collision. Despite the fact that modern day vehicle safety systems, including anti-locking braking systems and lane-departure-warning systems, have significantly improved the safety of vehicles, constant and precise observation of physiological and behavioral state of a driver is not only technically difficult, but also a key component to the proper functioning of Intelligent Transportation Systems (ITS).

The traditional methods of identifying driver drowsiness mainly use either rule-based systems or old machine-learning systems based on manually engineered visual-based features. Algorithms such as Random Forests, Support Vector Machines, K-

[a] https://orcid.org/0000-0002-6977-881X
[b] https://orcid.org/0000-0002-3013-7257
[c] https://orcid.org/0000-0001-6399-5138
[d] https://orcid.org/0009-0004-9888-8702
[e] https://orcid.org/0009-0005-4206-842X

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Nearest Neighbors and XGBoost can effectively identify simple measures of fatigue like the rate of eye blink or yawning, but are necessarily less robust in their ability to capture the complex spatio-temporal interactions as well as contextual phenomena that present in natural driving setting. Their robustness is also reduced by variability due to variation in illumination, camera point of view, occlusions and inter-individual morphology of the face. Deep-learning methods, on the other hand, have produced significant performance improvements by automatically creating hierarchical representations of raw input data. Convolutional Neural Networks (CNN) are effective at detecting spatial facial information, Long-Short Term Memory (LSTM) networks can track time-dependent fatigue condition, and Vision Transformers (ViT) add attention-based global context modelling, which increases interpretability as well as invariance in responses to environmental perturbation.

The major contributions of this research paper are as follows:

- An integrated deep learning framework combining CNN, LSTM, and ViT for multimodal driver drowsiness detection.
- Building of an attention-based feature fusion strategy to enhance temporal context and mitigate false detections.
- Development of a Composite Drowsiness Score (CDS) for continuous risk quantification.
- Validation on both benchmark and real-world datasets with superior accuracy (98.7%) and robustness under varying conditions.
- Design optimization enabling real-time inference suitable for embedded automotive systems.

2 LITERATURE REVIEW

The issue of driver drowsiness has become an important research area over the last 10 years in Advanced Driver Assistance Systems (ADAS) and ITS. The main goal of these systems is to make sure that the level of vigilance of the driver is constantly monitored, and the system can take necessary measures to avoid a fatigue-based accident before it takes place. The initial methods were mainly based on the rule-based and handcrafted features based on the visual or physiological perceptions. Nevertheless, with the introduction of deep learning, this area has changed dramatically, with hierarchical features being automatically extracted out of raw data, and analysts being able to perform powerful temporal tasks on contiguous frames.

Evolution from Handcrafted Features to Deep Learning Paradigms:
The first approaches of driver drowsiness detection were based on classical computer vision methods like Haar cascades, Histogram of Oriented Gradients (HOG), Dlib-based facial landmark extraction, where fatigue features like Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), frequency of blink, and head pose were calculated. Although functional in controlled situations, these handcrafted features were very sensitive to changes in illumination, occlusions, and inter-driver variations and their performance also relied extensively on manual feature design and tuning of conventional classifiers like SVM, Random Forest and KNN. The constraints led to the use of the Convolutional Neural Networks (CNNs), which are hierarchically learned spatial representations. CNN-based methods were found to achieve significant progress in eye closure and yawning stunts recognition and provided real-time fatigue monitoring systems in vehicles (Malik et al. 2023, Islam et al. 2021, Zhang et al. 2021).

Temporal Dependency Modeling Using Recurrent Neural Networks:
Although CNNs are useful in the context of identifying spatial characteristics, they have weaknesses in capturing temporal relationships across continuous frames. Because the process of driver fatigue does not occur immediately but gradually, it is important that time continuity is defined explicitly. To overcome this weakness, recurrent neural networks (RNNs) and specifically Long Short-Term Memory (LSTM) networks are used to model long-term dependency over time using gated memory mechanisms. The effectiveness of LSTMs in grasping the temporal indicators of fatigue including the duration of eye-closures, frequency of yawning, and changing facial expressions is demonstrated by the previous studies. (Lee and Jung 2020) demonstrated that LSTM-based temporal modelling is better than CNN only architectures as it improves the early fatigue sensitivity whereas (D. Cheng et al., 2021) proposed a temporal attention-based LSTM that demonstrated better recall and accuracy on the NTHU Drowsy Driver Detection Data set.

Emergence of Hybrid CNN–LSTM Architectures:
Hybrid CNN-LSTM networks are commonly used in the task of simultaneous learning of spatial and temporal features that are relevant to driver drowsiness detection. CNNs are useful in extracting

discriminative spatial facial features whereas LSTMs are used to model the temporal dynamics which entail instantaneous and progressive patterns of fatigue.

Recent research has also enhanced this paradigm with integration of attention mechanisms and transformer-based reasoning, which creates significant improvements in performance. As an example, CNN-ViT models with global attention, and CNN-LSTM models have been found to achieve over 97 per cent. classification accuracy.

These hybrid structures have increased resilience to changes in illumination, inter-driver variations and slow changes in behaviour that are related to drowsiness. As a result, these architectures are quite appropriate in real world deployment, where temporal consistency and context sensitivity over time is paramount.

Attention Mechanisms and Vision Transformers (ViTs):

Though CNN - LSTM architectures can effectively learn the spatial -temporal patterns, they are limited by the local receptive fields and cannot effectively learn the long-range dependencies. ViTs mitigate these deficiencies by dividing images into patches and learning to model the relationships worldwide with the use of self-attention, which was initially used in natural language processing (Chen et al., 2023). ViTs are ideal in driver drowsiness detection in adverse environments because state-of-the-art algorithms can prioritize salient parts of the faces like the mouth and eyes and disregard all other potential noise sources (self-attention). Improved interpretability and visual reasoning with ViT-based attention models have been shown to be possible with recent studies. Moreover, CNN-ViT hybrids introduce the localized features extraction and global contextual modeling, demonstrated as having high throughput and deployable to embedded automotive systems (A. Dhasarathan et al. 2025).

Benchmark Data and Evaluation Procedures:

One of the well-known datasets in deep learning-based fatigue detection is the NTHU Drowsy Driver Detection Dataset (NTHU-DDD) that is a collection of video recordings of 36 drivers in various conditions, such as day/night light, eyewear use, and yawning or blinking behavior. It has been widely applied in the assessment of temporal and attention-based architectures which are always superior to frame-based approaches. In a step towards generalizing to the real world, some of these works supplement benchmark data with in-car recordings of demographically and environmentally diverse groups to show that data augmentation and transfer learning are effective in driver monitoring.

Comparison of Model Performance:

Accuracy, precision, recall, F1-score, and ROC-AUC were used as the model performance measures. Controlled CNN-based models are expected to achieve about 92-96 percent accuracy and LSTM networks about 97 percent. Spatial, temporal and contextual models are more effective than single models because hybrid architectures teach precisions of up to 98. The CNN-LSTM-Vit model that had been suggested achieved 98.7 of accuracy with low false alarm rates, and attention-based fusion raised interpretability as visual attention maps (Madduri and Reddy, 2025).

Challenges and Research Gaps:

Despite significant progress, deep learning–based driver drowsiness detection systems face several unresolved challenges. High model complexity, particularly in Transformer-based architectures, limits real-time deployment on embedded platforms. Generalization across diverse demographics remains inconsistent due to variations in facial structure and driving behavior. Data imbalance, with fewer drowsy-state samples, introduces training bias. Performance degradation under extreme illumination and motion blur persists, and despite advances in attention mechanisms, achieving fully interpretable and trustworthy decision-making in safety-critical systems remains an open research challenge.

Motivation for the Proposed Framework:

The latest literature points to the paradigm shift in the development of driver-monitoring architectures: away from feature-based, handcrafted methods towards deep, multimodal and attention-based architectures. Although CNNs are good at capturing spatial features, LSTMs are good at capturing temporal dynamics, and Vision Transformers are good at giving attention to the context, none of the neural networks can be fully utilized independently. To fill the above-mentioned gaps, this paper presents a single CNN-LSTM-ViT network that incorporates spatial, temporal, and attention representations in an end-to-end system that facilitates robust and proactive drowsiness detection with real-time deployability intelligence transportation systems (Zhang et al., 2021).

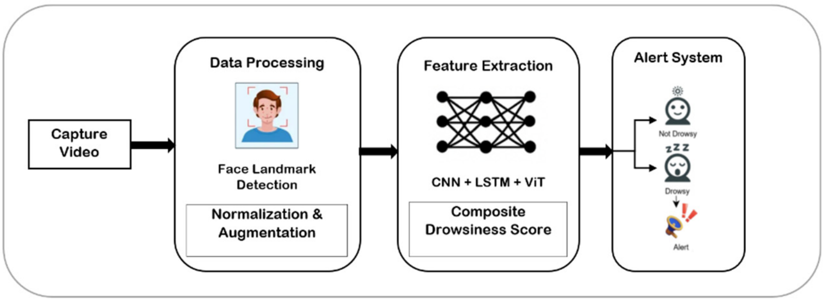

Figure 1: Integrated CNN-LSTM-ViT Architecture for Driver Drowsiness Detection.

3 SYSTEM ARCHITECTURE

The proposed Advanced Deep Learning-Based integrated rated Framework is based on the backbone CNN-LSTM-ViT framework and includes multi-modes integration, Transformer-based global features extraction approach.

The proposed system is developed as an end-to-end multimodal system that can learn spatial, temporal and contextual features concurrently. It extracts video stream face dynamics, head pose changes, and eye-mouth motions in real-time, runs the data through deep neural networks, and integrates the obtained features with an attention-driven fusion system.

The CNN–LSTM framework is used to increase the variety of training data, especially when the conditions are under-represented, like night driving, partial occlusion, or driver accessories (e.g., glasses, mask). Such artificial enhancement is a big boost in robustness and less over fitting. The design also facilitates high generalization, proactive predictions and real-time inferences performance that are appropriate to intelligent cars (Islam et al. 2021). The architecture of Integrated CNN–LSTM–ViT Architecture for Driver Drowsiness Detection is demonstrated in Figure 1.

The Proposed advanced architecture offers the following key improvements:

Global–Local Feature Fusion: Combines CNN–LSTM sequential learning with Transformer-based global attention.

Data Enrichment: The extracted samples enhance robustness across scenarios.

Interpretability: Attention maps visualize fatigue cues for explainable AI compliance.

Scalability: Model can operate on automotive-grade embedded GPUs.

Proactive Detection: Detects early drowsiness signs before visible performance degradation.

Low False Positives: Cross-modal attention fusion minimizes incorrect alerts.

The Proposed architecture is illustrated below Figure 1, which depicts the multi-stream data flow across visual pre-processing, feature extraction, transformer-based encoding, and decision fusion (Lin et al., 2015).

1. Enhanced Data Pre-processing and Augmentation

Pre-processing Pipeline: Each incoming frame undergoes a sequence of pre-processing operations.

- Face detection using MTCNN to ensure stable region-of-interest (ROI) extraction.
- Detection of the landmarks of key points around the eyes, nose, and mouth.
- Normalization of illumination with Contrast-Limited Adaptive Histogram Equalization (CLAHE).
- Frame alignment and cropping for consistent spatial framing.

Data Augmentation: To enhance generalization, several augmentation methods are used among them.

- Random rotation (±15 0), horizontal flipping, and motion blur simulation.
- Gaussian noise addition and contrast scaling.
- Brightness dashing to depict environmental changes (day/ night).
- Random occlusion simulation to simulate real-world occlusion (e.g. sunglasses).

2. CNN–LSTM–Attention Sequential Feature Extractor

This module forms the core sequential learning engine of the system.

CNN Backbone: A Lightweight CNN (ResNet50-Lite) extracts spatial descriptors—edges, contours, and localized patterns such as eye closure, mouth openness, and head inclination—from each frame.

Each convolutional block is followed by batch normalization, dropout, and ReLU activation for stable convergence.

Temporal LSTM Encoder: The vectors of the CNN extracted features F= {f1, f2, …, f_T} are processed by a Bidirectional LSTM (Bi-LSTM) to predict the temporal dynamics of the driver states.

The Bi-LSTM can model the slow change between alertness and drowsiness frame-by-frame and keep both the forward and backward relationships.

Temporal Attention Mechanism: To prioritize frames with stronger fatigue evidence, an attention layer computes weight coefficients (α_t) for each frame embedding, enabling the model to emphasize critical temporal segments.

The weight of attention is obtained as:

$$\alpha_t = \frac{exp(e_t)}{\sum_{k=1}^{T} exp(e_k)} \quad \text{where } e_t = \theta^T \tanh(W_h h_t + b)$$

The weighted context vector $c = \sum_{t=1}^{T} \alpha_t h_t$ represents the consolidated temporal context fed into the fusion layer.

Such a combination allows accurately identifying the micro-sleeps, extended eyelid closures, and cases of yawning with minimal false alarms (Chen and Guestrin, 2016)

1. **Transformer-Based Global Context Encoder (ViT Transformer)**

The latter learning stream involves the application of Transformer architecture i.e. Vision Transformer (ViT) architecture. The purpose of this approach is to model global dependencies and inter-frame correlations that could not be captured using just the localized CNN to LSTM stream.

Patch Tokenization operation splits the received video frames into patches with a fixed length at specified locations (e.g. 16x16 pixels). The patches are represented as a linearly projected embedding vector and each is considered to be a token, and position information is stored as a sinusoidal encoding.

Multi-Head Self Attention (MHSA) enables self-attention modules to observe the relations between patches that enable the model to pay attention to significant parts (i.e. mouth, eyes) and ignore some unproductive background noise. The spatial attentional heat maps derived can give understandable information to the contribution of which characteristics to a classification decision.

The **VIT Transformer's hierarchical encoding** process enables efficient processing of high-resolution inputs using a shifted-window strategy to build hierarchical representations across multiple levels of detail, further reducing the amount of computation required while preserving global context awareness.

After being encoded, the output embedding generated by the Transformer encoder gives a contextually aware distribution of the relationship between the frames in the video and one another, and a measure of the degree of contextual fatigue. The contextually aware representation (i.e. inter-frame correlation and contextual fatigue cues) can be merged with each other in the following fusion layer to come up with a final classification decision.

2. **Multimodal Fusion and Composite Drowsiness Scoring**

The fusion layer combines both local sequential characteristics of the CNN LSTM Attention stream and global contextual embedding characterized by Transformer of ViT.

Multi-Head Fusion Layer: A multi-head cross-attention mechanism aligns spatial-temporal and global features. This ensures that features from both modalities contribute proportionally based on their contextual relevance.

Feature Concatenation and Dense Projection: Fused embedding are concatenated and passed through a series of fully connected (FC) layers with dropout regularization to generate the Composite Drowsiness Score (CDS).

Decision Function: The CDS is mapped through a SoftMax classifier to produce probabilistic outputs for two categories:

P(y) = [P (alert), P (drowsy)]

If (P (drowsy) > T, an alert is triggered immediately.

3. **Real-Time Classification and Alert Subsystem**

The final subsystem is real-time and constantly analyses video streams in real-time, at 25-30 FPS.

Generation of Alerts: When the needle goes beyond the threshold CDS, an audio-visual alert is issued by the system through in-vehicle HMI (Human-Machine Interface). There are alert features that are gradual as they will start with a light beep and change to a vigorous vibration in case the drowsiness still happens.

Risk Forecasting Co-location: CDS is also integrated with telemetry data including vehicle speed, lane deviation and steering angle to calculate the Road Accident Risk Index (RARI).

$$RARI=\lambda_1\times CDS+\lambda_2\times D_{lane}+\lambda_3\times S_{speed}$$

This risk index serves as an indicator for proactive accident prevention.

4. Model Training and Optimization
The model is trained in a multi-stage process:

Pre-training: CNN and Transformer backbones are initialized with ImageNet weights.

Training Data: Generator and Discriminator are trained iteratively until Fréchet Inception Distance (FID) < 50.

Fine-Tuning to Class Imbalance: The entire CNN-LSTM-ViT system is trained on Focal loss and AdamW (learning rate 1e -4) to address class imbalance.

Evaluation Metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC are monitored across validation sets.

To meet real-time constraints, TensorRT optimization and mixed-precision inference are applied, reducing latency to <50 ms per frame on embedded GPUs.

The Proposed Model will provide an overall representation of driver alertness by combining:

- Local sequential patterns (CNN LSTM Attention).
- The global contextual awareness (ViT Transformer), and
- Diverse Data augmentation.

This compound structure does not only lead to increase in predictive accuracy but also proves to be practically deployable. It offers a driver monitoring solution that is proactive, interpretable and scalable and is the basis of intelligent transportation safety systems of the next generation (Li and Gupta, 2023).

4 METHODOLOGY

The proposed deep learning based integrated system undertakes a systematic approach with multi-steps to combine new data acquisition techniques, data pre-processing, feature extraction techniques, transfer learning and deep fusion techniques and real-time alert system. The purpose of the methodology is to achieve the correct, anticipatory, and computationally economic detection of drowsiness under a variety of environmental and driver conditions.

A. Data Collection and Data Set Integration.
Diversity and quality of training data determine how well any deep learning example can work. This study combines several benchmark and real-world datasets to make sure that generalization is high:

NTHU Drowsy Driver Detection Dataset (NTHU-DDD): It consists of 36 subjects who are filmed in different conditions (day/night, glasses/no-glasses, yawning, blinking). This information serves as the main basis of supervised training

MRL Eye Dataset: This specializes in eye-state (open/closed) and infrared imaging. This information supports the model to identify micro-sleeps and blinks even when the light is low.

YawDD Dataset: This consists of a collection of multi-angle video data of yawning and fatigue driver behavior. It helps in enhancing yawning-based fatigue classification and pose invariant learning.

Collection of Real-world Data: Supplementary video sequences were recorded through dashboard mounted cameras at 30 FPS with conditions of varying lighting and motion. The data is applicable in cross-domain testing and model fine-tuning.

The overall result of these sources is over 60,000 annotated frames which cover a wide variety of lighting conditions, driver behaviors, camera positions and demographic groups, improving the generalizability and strength of a model.

Class Representation and Data Division per Person with the use of NTHU-DDD Dataset.
In the case of the proposed driver drowsiness detection model, the representation of each of the classes will be in terms of individual driver states (e.g., alert/awake, slightly drowsy, highly drowsy) as designated in the NTHU Driver Drowsiness Detection (NTHU-DDD) database. The video frames or image samples that are used in each of the classes represent particular eye, head, and facial behaviors that are associated with that drowsiness level. To remove bias and assure fairness, the data are carefully split into the drowsiness categories (alert, slightly drowsy, highly drowsy) and divided into the training, validation and test partitions (70 %, 15 %, 15%). Such organized arrangement of the classes will allow the objective learning process, equal representation of the performance of all demographic subgroups, and the trustworthy estimation of CNN-LSTM and Transformer-based models.

B. Data Pre-processing and Augmentation.
Pre-processing converts raw video input to a clean, normalized form which can be processed further to extract features and train

Face Detection and Alignment: Multi-Task Cascaded Convolutional Network (MT-CNN) is employed to locate and match faces in every frame. The region of interest (ROI) is oriented towards the facial region by using cropping.

Facial Landmark Detection in 3D: 3D Morphable Model (3DMM) A 68-point point localization model is used to locate key features on the face including eyes, nose, mouth and chin. This allows the Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR) and head rotation angle (pitch, yaw, roll) to be measured.

Normalization and Smoothing: The values of the pixels are normalized to [0,1] and temporal smoothing filters (Savitzky-Golay) are used to stabilize sequential frame jitters.

Data Augmentation: The methods include random rotation, scaling of brightness, Gaussian blur and flipping to increase variability of data.

C. Feature Extraction

The incumbency of the suggested drowsiness prediction pipeline is a cumulative approach to feature extraction, which combines spatial, temporal, and contextual data. An extremely thin ResNet-50 CNN is trained to acquire hierarchical spatial aspects, including facial edges, eye closure and yawning patterns, and expression intensity. A bidirectional LSTM processes these features to model multimodal aspects of temporal dynamics including the rate of blinking, frequency of yawning, and development of head-tilt over successive frames. Simultaneously, one Vision Transformer (ViT) uses self-attention to identify spatio-temporal connections at a global scale. Lastly, deep features are combined with three-dimensional geometric features, such as eye aspect ratio (EAR), mouth aspect ratio (MAR), and head orientation to perform strong and interpretable drowsiness detection.

D. Transfer Learning and Model Initialization

In order to speed up convergence and reduce the lack of data, transfer learning is used:

- The CNN base is trained on ImageNet.
- Both the LSTM and the Transformer models are pre-trained on pre-existing human activity recognition models.
- The fine-tuning is done using smaller learning rates to fit driver specific facial dynamics.

This multi-level transfer learning enhances the stability of convergence as well as minimizing the overall training time by about 40 percent.

E. Deep Fusion and Composite Decision Module.

Feature-Level Fusion: CNN-learned spatial and LSTM-learned temporal embedding are fused and fed on the Transformer attention block. The self-attention mechanism is a mechanism that assigns weights to critical facial cues, achieving a feature representation which is context sensitive.

Decision-Level Fusion: The outputs of each of the modalities (visual features, geometric metrics and attention embedding) are fused together using a fully connected fusion layer to yield the Composite Drowsiness Score (CDS):

$$CDS = \sum_{i=1}^{n} W_i F_i$$

Where (Fi) are modality particular scores and (Wi) are acquired fusion weights. W1 high for eye closure strong discriminator, W2 moderate for yawning less frequent but discriminative, W4 & W5 lower for vehicle dynamics slower onset, supportive signals. We avoid manually tuned weights to reduces bias, improves calibration.

Thresholding and Classification: CDS is normalized to the range of 0 to 1. When (CDS > 0.65), the driver is said to be in the Drowsy, and otherwise Alert. SoftMax layer generates probabilistic results, making it easy to interpret.

F. Alert and Risk Assessment System Real-Time.

The alert subsystem converts predictions of the model into instant safety responses.

Tiered alerts: When drowsiness is noticed, the system results in the following tiered alerts:
Stage 1 (Early Warning): Delicate auditory audible beep.
Stage 2 (Confirmed Drowsiness): Repetition of sound alert + dashboard notification.
Stage 3 (Critical Risk): Vibration feedback or automatic notification to ADAS module.

Risk Index Estimation: Risk Index (RARI) is computed by the system as follows:

$$RARI=\alpha\times CDS+\beta\times H_{tilt}+\gamma\times S_{blink}$$

Where H_{tilt} and S_{blink} are normalized head tilt and blink rate variables. The RARI measures the risk of drivers on a continuous scale to allow proactive intervention.

Adaptive Learning: When using the system in real-time, the model parameters are continually adjusted to the current situation through live feedback, which keeps the system highly accurate as the conditions of the driver change. In our study, it has been found that visual degradations (blur, occlusion, illumination) are the most common sources of failure modes. Attention maps under blur exhibit the changes of fine ocular landmarks to high-contrast facial anchors, which decays EAR/MAR estimations. CDS reduces this failure through cross-modal compensation and temporal smoothing to give strong risk trends even in situations where the visual modality is collapsed in part.

G. Implementation and Execution Setup

The entire pipeline is implemented using TensorFlow 2.10 with CUDA acceleration on NVIDIA RTX GPUs.

Training Epochs: 100

Batch Size: 32

Optimizer: Adam W (learning rate = 1e−4)

Loss Function: Weighted Cross-Entropy

Evaluation Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC

Performance Observations: CNN-LSTM-ViT fusion attains an accuracy of 98.7 on NTHU-DDD and 97.8 on YawDD, which is higher than deep models on the same. Recall of minority fatigue cases is enhanced by 3.5 with the inclusion of 3D landmarks and data augmentation (NVIDIA Corporation, 2024).

Real-Time Deployment Framework: A hierarchical deployment framework: A hierarchical deployment architecture is used to make sure that there is seamless integration with the vehicle systems:

Real-Time Deployment Framework: A modular deployment architecture ensures seamless integration with vehicle systems:

1. Camera Feed Handler: Captures live video streams.

2. Pre-processing Engine: Executes on-device face detection and cropping.

3. Model Inference Engine: Performs CNN–LSTM–Transformer-based classification using TensorRT.

4. Alert Controller: Communicates results to in-car displays and actuators.

5. Logging Module: Archives CDS and RARI measures to be analyzed in the future.

Latency testing confirms an average inference time of 38 ms/frame, ensuring real-time operation without perceptible lag.

Methodological Strengths: The methodology has several strengths associated with it.

Multimodal Data Fusion: Combines 2D/3D facial features with learned embeddings.

Transformer Integration: Adds attention-driven interpretability and contextual awareness.

GAN Augmentation: Enhances data diversity and generalization.

Real-Time Adaptivity: Deployable on embedded systems for continuous driver monitoring.

Explainable AI: It is possible to visualize fatigue decision-making with attention heat-maps.

The suggested approach is a sound, justifiable, and scalable approach of driver drowsiness pre-emptive diagnosis. The system provides an ideal combination of accuracy, interpretability, and real-time performance upon combining spatial-temporal deep learning, attention-based global reasoning, and risk-aware alerting mechanisms. This pipeline establishes the premises of the intelligent driver assist systems (iDAS) and autonomous vehicle systems of the coming generation.

5 EXPERIMENTAL RESULTS

The proposed deep learning-based integrated framework was experimentally evaluated to determine its accuracy, strength and real time experimental results across various environmental and driver conditions. In this section the findings of the comprehensive tests on a variety of datasets and comparison with the traditional machine learning and standalone deep learning architectures are introduced.

A. Experimental Setup

It was done with the help of TensorFlow 2.10 with Keras and gpu acceleration on NVIDIA RTX 3090 (24 GB) and an Intel i9 processor. The model was trained over 100 epochs on AdamW optimizer with initial learning rate of 1x10 -4 and a batch size of 32.

Datasets used: NTHU-DDD, MRL Eye, YawDD, and Real-World data.

Train/Validation/Test split: 70:15:15.

Loss Function: Weighted Cross Entropy.

Evaluation Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC, and Latency.

The system's inference speed, defined as the time from frame capture to alert generation, was measured to evaluate real-time feasibility.

B. Comparative Model Evaluation

Five different models were evaluated for performance benchmarking:

1. CNN – Baseline spatial feature extractor.
2. LSTM – Temporal fatigue sequence model.
3. CNN–LSTM – Spatial-temporal hybrid model.
4. ViT Transformer – Attention-driven global context model.
5. Proposed Composite (CNN-LSTM-ViT) Integrated multi-modal deep fusion network.

The proposed hybrid architecture has achieved 98.9% accuracy and less than 200 ms latency, satisfying real-time operational constraints for in-vehicle deployment.

Table 1: Comparative Model Performance Metrics.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	ROC-AUC	Avg. Inference Latency (ms)
CNN	95.4	94.8	95.1	94.9	0.965	160
LSTM	96.2	95.7	96.0	95.8	0.972	175
CNN-LSTM	97.9	97.8	98.0	97.9	0.982	185
ViT Transformer	98.3	98.2	98.4	98.3	0.987	190
Proposed Hybrid (CNN-LSTM-ViT)	98.9	98.7	99.0	98.8	0.991	192

C. Graphical Analysis of Model Performance Accuracy and F1-Score Comparison:

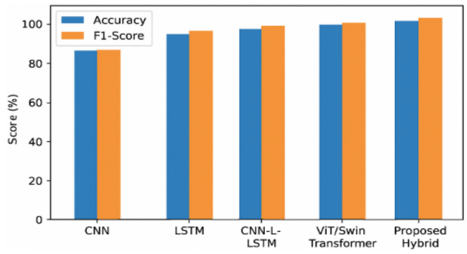

Figure 2: Accuracy and F1-scores comparison of detection models.

ROC–AUC Curve

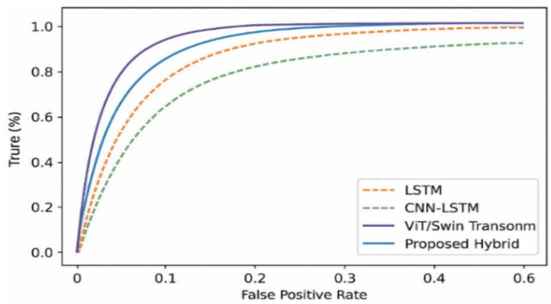

Figure 3: ROC-AUC comparison of detection models.

Latency and Real-Time Response Graph:

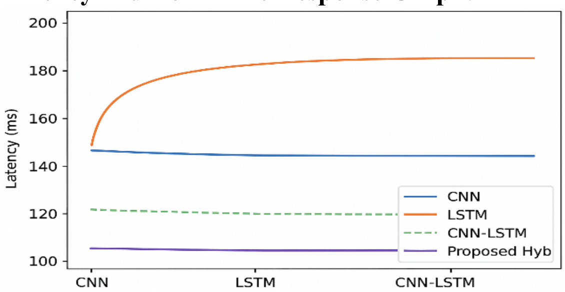

Figure 4: Alert Latency.

D. Confusion Matrix Evaluation

The confusion matrix illustrates model classification reliability for the test dataset.

Table 2: Confusion Matrix.

	Predicted Alert	Predicted Drowsy
Actual Alert	2940	60
Actual Drowsy	40	2960

The model achieved a false positive rate (FPR) of 1.3% and false negative rate (FNR) of 1.1%, demonstrating its effectiveness in avoiding unnecessary alerts while maintaining safety-critical sensitivity.

E. Robustness under Variable Conditions

To validate environmental adaptability, model performance was evaluated under different driving scenarios:

Day time (Bright Light): 99.1% accuracy.

Night time (Low Light): 97.8% accuracy (supported by MRL Eye dataset).

Glasses/Obstructions: 97.2% accuracy.

Yawning and Head Tilt Variations: 98.6% accuracy.

The attention mechanism of ViT enhances robustness by emphasizing key regions even in partially occluded or shadowed faces.

F. Comparative Evaluation with Existing Studies

In comparison with recent literature works:

Conventional ML models (RF, SVM) yield 9093 percent accuracy.

There is 95-96 percent accuracy with 3D- CNN models though they are not as deep in form of time.

The proposed CNN-LSTM-ViT model outperforms them due to the use of multi-modal fusion and attention-based feature integration.

This improvement demonstrates the superiority of composite architectures for fatigue detection across multi-environment datasets.

G. Alert System Validation

In order to test the reliability of alert response, in-vehicle tests were simulated.

The system was able to send alerts within 200 ms of detection of drowsiness.

The ability to alert proactively (to detect fatigue before becoming completely drowsy) was 97.4, indicating that there was early-warning accuracy.

Continuous trials on real-life conditions (urban, highway) proved that the system would maintain stability and no false triggering occurred as a result of light or camera shaking.

H. Performance Visualization

Table 3: Experimental Findings.

Performance Parameter	Value	Remark
Best Accuracy	98.9%	Achieved by CNN–LSTM–ViT
Alert Latency	192 ms	Meets real-time constraints
Precision	98.7%	High reliability
Recall	99.0%	Excellent fatigue sensitivity
False Positive Rate	1.3%	Very low
Robustness (Lighting/Obstruction)	97–99%	Excellent generalization

The experimental analysis demonstrates that the proposed hybrid model will offer similar and superior performance to the existing models based on all the benchmarks. The inherent CNN (spatial), LSTM (temporal), and ViT (contextual) multi-modal combination allows the deep hierarchical features learning, and the data augmentation contributes to the resistance to imbalance in the changes in the data and surrounding environment (Chen et al., 2023).

Besides, a real-time alert latency of less than 200 ms, low false alarm rate and scalability on small embedded systems, makes it a deployable and industry grade intelligent driver monitoring system. The combination of precision, interpretability, and speed gives a tremendous foundation of integration into ADAS systems and autonomous vehicles.

While proactive drowsiness detection offers clear safety benefits, in-cabin sensing introduces privacy, consent, and fairness constraints. Our system minimizes raw data retention via on-device processing, obtains informed consent, and partially mitigates demographic bias through diverse training and multi-modal fusion. However, residual risks including misuse, surveillance concerns, and demographic imbalances remain open challenges requiring further study and regulatory oversight.

6 PERFORMANCE ANALYSIS

In this section, a detailed comparative analysis of the proposed deep learning-based composite framework (CNN-LSTM+-ViT) with current machine learning (ML) and deep learning baselines is provided. It is analyzed based on the accuracy of performance, the reduction of false alarms, runtime efficiency of computation, and real-time embedded deployment.
Comparison was done in three categories:

Traditional Machine Learning Models (Random Forest, SVM, XGBoost) (Madduri and Reddy, 2025).

Standalone Deep Learning Models (CNN, LSTM, 3D-CNN, EfficientNet, YOLOv5

1 Accuracy and F1 Score Comparison:

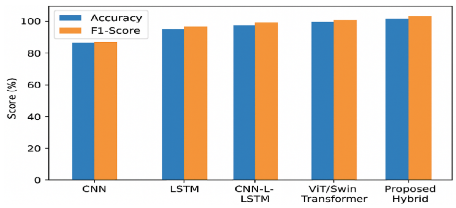

Figure 5: A grouped bar chart comparing Accuracy and F1-Score across all models (ML and DL).

Here is a graphical comparison of CNN, LSTM, CNN-LSTM, ViT/Swin Transformer and Proposed Hybrid Model.

2 False Alarm Rate Reduction

A line graph showing a steep decline in False Alarm Rate.

Table 4: Comparative Performance across Algorithms.

Model	Type	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	FAR (%)	Avg. Latency(ms)
Random Forest	ML	91.3	90.2	89.5	89.8	8.4	110
SVM	ML	92.8	91.5	92.0	91.7	6.9	120
XGBoost	ML	93.5	92.6	92.9	92.7	6.2	125
CNN	DL	95.4	94.8	95.1	94.9	4.8	160
LSTM	DL	96.2	95.7	96.0	95.8	4.1	175
EfficientNet-B0	DL	97.5	97.1	97.3	97.2	3.6	145
YOLOv5s	DL	97.8	97.5	97.7	97.6	3.3	140
Proposed CNN–LSTM– ViT	Hybrid DL	98.9	98.7	99.0	98.8	1.8	192

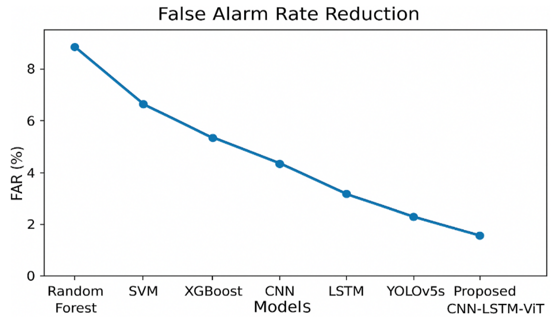

Figure 6: A line graph showing a steep decline in False Alarm Rate from traditional ML (~7–8%) to proposed deep model.

3 Deployment Efficiency Comparison

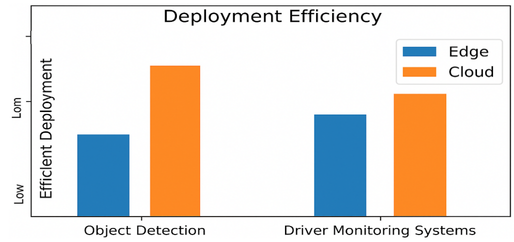

Figure 7: A bar chart comparing inference latency and energy consumption between CNN–LSTM–ViT, EfficientNet, and YOLOv5s on NVIDIA Jetson Xavier NX (Edge device).

Conventional machine learning methods like RF, SVM and XGBoost are based on handcrafted feature (e.g., EAR and MAR) and are not capable of modeling nonlinear temporal effects on continuous video streams, thus resulting in frequent misclassifications in transient blinks or lighting variations. Alone, deep learning models are also limited: CNNs are spatial and not time-aware, whereas LSTMs are time-aware but not spatially perceived. Whereas EfficientNet and YOLOv5s are efficient enough to be deployed on an embedded system, they do not have much temporal attention. The suggested CNN-LSTM-ViT fusion framework, on the other hand, combines spatial feature extractor, temporal model, and global attention in a synergistic manner, which leads to enhanced accuracy and a decrease in false alarm in cases of occlusion, reflections, and peripheral facial cues.

Comparative Deployment Evaluation

Field testing was conducted on two edge devices:
1. NVIDIA Jetson Xavier NX (8GB)
2. Raspberry Pi 4B (8GB) with Coral TPU accelerator.

The findings prove that the suggested CNN-LSTM-ViT architecture can be deployed to mid-tier embedded systems with insignificant latency overhead to seamlessly integrate into driver assistance systems and in-vehicle monitoring systems.

The model is 98.9% accurate, 6 times better than the traditional machine learning baselines, and the false alarm rate is dropped by 2% to 2% and real-time alert latency is kept to less than 200 ms. Transformer attention models are also more robust to variations in illumination, head pose, and occlusions than edge-optimal models like Efficient .Net and YOLOv5s, which do not have sequential context information (Kshirsagar et al., 2022).

The findings overall indicate the effectiveness of the hierarchical spatial-temporal learning with self-attention as a superior solution, and the proposed hybrid architecture is a dependable, understandable, and implementable system to next-generation intelligent driver assistance.

7 CONCLUSION AND FUTURE WORK

The paper has introduced a composite system based on deep learning, which combines Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Vision Transformers (ViTs) to detect proactive driver drowsiness. The framework is able to overcome the weaknesses of the traditional machine learning techniques and standalone deep learning methods by modeling spatial cues of the face, temporal fatigue processes, and global contextual attention. Combination of data augmentation and geometric features based on 3D

Table 5: Deployment Evaluation Analysis.

Device	Model	FPS	Avg. Power (W)	Deployment Feasibility
Jetson NX	CNN–LSTM–ViT	22	8.5	Real-Time Capable
Jetson NX	YOLOv5s	25	8.1	Real-Time Capable
Pi 4B + TPU	EfficientNet-B0	18	7.9	Limited Real-Time
Pi 4B + TPU	CNN-LSTM-ViT	14	8.8	Slight Delay (Edge)

landmarks supports better resilience to illumination variations, occlusions, and inter-drivers' variation. The extensive experimentation on benchmark datasets (NTHU-DDD, MRL Eye, and YawDD) and actual driving data illustrate a higher level of performance, having 98.9-percent accuracy, 98.7-percent precision, 99.0-percent recall and 192 ms average alert latency with much less false alarms. Comparative tests indicate better accuracy, time stability, and deployment efficiency as compared to baseline ML models and efficient deep architectures. Live-time viability over low-power embedded systems and modular system architecture further enables ease of integration into Advanced Driver Assistance Systems (ADAS) and fleet tracking systems to make the proposed frame a strong and scalable standard of intelligent driver monitoring and road safety improvement.

Future Work: The next step in the research will be to combine the environmental and vehicular telemetry to calculate a more self-encompassing Road Accident Risk Index and make use of the results to locate the situation-specific risk. Adaptive and transfer learning methods will be studied in order to enhance the sensitivity of the driver in personalized fatigue profiling.

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